Test-time Scaling for Diffusion Models using Generative Uncertainty

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Abstract

Inference-time scaling improves diffusion models by searching over the initial noise with an external verifier, at the cost of verifier bias and the need for curated preference data. Generative uncertainty offers an internal alternative: it scores a generation by the semantic disagreement among M completions of the same noise under a Laplace posterior over the network weights. So far it has only been used as a post-hoc filter, because it requires M sequential denoising runs per candidate, which is prohibitive inside a search loop. This thesis makes scoring search-ready in two steps. First, we replace the weight perturbations with perturbations of the initial noise itself, so that the M completions share one batched forward pass. On class-conditional ImageNet 256×256 with a U-ViT-H/2 backbone, filtering a fixed pool by this score matches the original Laplace filter at the FID optimum (8.86 vs. 8.76; unfiltered 10.66). Second, we used the score as the objective of a zero-order search over initial noise at 400 NFE per image. Requiring no external model, no labels, and no preference data, the search clearly beats unguided sampling (FID 7.99 vs. 10.66) and preserves the most sample diversity among all search objectives tested, but it is outperformed at matched compute by both a labelled and a label-free external CLIP verifier (6.95 and 7.26). On this benchmark, the model's own uncertainty is a usable, verifier-free search signal, yet a weaker one than an external CLIP verifier.

Keywords: diffusion models, inference-time scaling, test-time compute, generative uncertainty, Laplace approximation, zero-order search, verifier-free search

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1 Introduction

Modern image generators are neural networks that learn from millions of example photographs what natural images look like, and can then synthesise new images on demand. The quality of what they produce depends not only on how well they were trained but also on how much computation is spent when an image is generated and on how that computation is spent.

The dominant such generators today are denoising diffusion models [8], which have displaced generative adversarial networks [6] and variational autoencoders [10] as the standard framework for image generation on benchmarks such as ImageNet [5] and COCO [14]. The model is trained to invert a fixed process that gradually corrupts the data with Gaussian noise. At generation time, a latent $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is denoised step by step, optionally conditioned on a class label or text embedding, until a clean image emerges. Each step is a forward pass through a large network, so the cost of generation is counted in *neural function evaluations* (NFE).

An NFE budget can be spent along two axes. The classical axis is the number of *denoising steps* spent on a single sample. Initially, this axis was expensive: the original DDPM sampler [8] is stochastic and needs hundreds to a thousand steps at one NFE each. Deterministic samplers that instead solve the probability-flow ODE, from first-order DDIM [20] to higher-order solvers such as DPM-Solver [15], cut the steps needed for a given quality to a few tens. a faster sampler only lowers the cost of reaching that quality. It does not raise the quality that more steps can reach. Once the step count is moderate, additional steps on the same sample barely improve it [16], so quality cannot be bought indefinitely along this axis.

The second axis is imported from language modelling, where sampling many candidate answers and selecting among them improves accuracy predictably with the number of candidates [2, 19]. Transferred to diffusion [16], it rests on a single observation. Under a deterministic sampler the generated image is a fixed function of the initial noise. Not all noise is equally good: some latents decode into clean, on-distribution images and others do not. Inference-time scaling thus becomes a *search over initial noise*, guided by a *verifier* that scores each candidate generation. Searching with verifiers such as CLIPScore, aesthetic predictors, or human-preference reward models yields large gains in FID and human preference [16].

The verifier is the bottleneck of this recipe. It is an *external* model that must be trained or chosen for the metric of interest, and it imports its own biases: the search optimises the verifier's proxy rather than sample quality itself, an effect documented as "verifier hacking" [16]. A signal that is *internal* to the diffusion model, requires no preference data, and commits to no particular notion of quality would remove this dependency.

Such signals already exist: Bayesian treatments of diffusion models yield per-sample uncertainty estimates for generations [11, 9]. This thesis builds on one of them, *generative uncertainty* [9]. A last-layer Laplace approximation [4] places an approximate posterior over the network weights. Drawing M weight samples, completing the *same* $\mathbf{x}_T$ under each, and measuring the semantic disagreement of the completions in CLIP feature space [17] yields the score. Low disagreement marks a generation the model is confident about, high disagreement a fragile one. Crucially, the method was proposed and validated only as a *post-hoc filter*: generate a pool of samples, discard the most uncertain, and the retained set has markedly lower FID than heuristic baselines. It was never used to *search*, and its cost explains why. Each of the M completions carries a different weight vector, so the M sampler runs cannot share a batched forward pass: on a single GPU the score costs M *sequential* full trajectories.

This thesis turns generative uncertainty into a search objective in two steps. First, we replace the weight-space perturbations with perturbations of the initial noise itself. All M completions then share one frozen set of weights, collapse into a single batched forward pass, and may be rendered on a shorter denoising schedule than the kept image. A score that was sequential in M thus becomes parallel in M. Second, we place this parallelised score inside a zero-order search over $\mathbf{x}_T$ [16], where it acts as a *verifier-free* search objective, and compare it head-to-head against an external CLIP verifier at matched inference compute.

We study class-conditional generation on ImageNet-1k 256×256 [5] with a U-ViT-H/2 backbone [1] in the latent space of a pretrained autoencoder [18], sampled deterministically with the second-order DPM-Solver [15]. Within this setting the central question is

> *Can the model's own generative uncertainty replace an external verifier, both as a score for selecting samples and as an objective for searching over initial noise, so that the same inference compute yields samples closer to the data distribution?*

We decompose this into two research questions.

RQ1: Can input-noise perturbations replace the Laplace posterior as the source of perturbations in generative uncertainty? We keep the original generative-uncertainty estimator [9] intact: M perturbed completions, a CLIP embedding, and the entropy of the induced Gaussian. We change only *where* the perturbation enters: Gaussian noise on $\mathbf{x}_T$ instead of noise on the last-layer weights. RQ1 asks whether this substitution is empirically safe: does filtering a fixed pool by the input-perturbation score retain samples with FID and Improved Precision and Recall [12] comparable to filtering by the Laplace score? A positive answer yields a parallelisable estimator and makes RQ2 tractable.

RQ2: Can generative uncertainty serve as the verifier in a search over initial noise, and how does it compare to an external verifier at matched compute? Using the estimator validated in RQ1 as the objective, we run zero-order search over $\mathbf{x}_T$: from a random pivot, repeatedly propose neighbouring latents, keep the one of lowest uncertainty, and render the final pivot at full quality. At a fixed total NFE budget we compare this against the identical search loop driven by an external CLIP verifier [16]. The CLIP verifier rewards images that match CLIP's notion of the target class. Generative uncertainty only asks whether the model agrees with itself under perturbation. RQ2 asks which signal buys more sample quality, measured by FID and Improved Precision and Recall.

2 Background

2.1 Diffusion models

Generative modelling seeks to draw new samples from an unknown data distribution $q(\mathbf{x}_0)$ given a finite training set. Learning a direct map from noise to image is hard, and the GANs and VAEs that attempt it suffer from unstable optimisation or oversmoothed samples [6, 10]. Diffusion models [8] replace this single map with three stages. A fixed *forward process* gradually corrupts each image into Gaussian noise through a sequence of small noising steps. A *denoiser* $\boldsymbol{\epsilon}_\theta$ is trained to invert one of such steps. A *sampler* then applies $\boldsymbol{\epsilon}_\theta$ repeatedly to transport the noise back to the data distribution. Because each step inverts only a small Gaussian perturbation, modelling $q(\mathbf{x}_0)$ reduces to a sequence of regression problems rather than a single generative leap. We treat the stages in turn and reach the premise on which this thesis is based: under a deterministic sampler, the generated image is determined entirely by the initial noise $\mathbf{x}_T$.

Forward process. The forward process is a Markov chain that adds Gaussian noise on a variance schedule $\beta_1, \ldots, \beta_T \in (0, 1)$,

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\Big(\mathbf{x}_t;\ \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t \mathbf{I}\Big). \tag{1}$$

Successive steps compose in closed form (with $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s \le t} \alpha_s$), giving the marginal

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{2}$$

The schedule drives $\bar{\alpha}_T \to 0$, so $\mathbf{x}_T$ is indistinguishable from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and generation can begin from pure noise.

Learning the reverse process. The network $\boldsymbol{\epsilon}_\theta$ is trained to predict the noise added to a corrupted image. Training pairs (input $\mathbf{x}_t$, target $\boldsymbol{\epsilon}$) follow from Equation (2), which reduces the objective to a regression,

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\,\mathbf{x}_0,\,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\ t\big)\right\|_2^2\right]. \tag{3}$$

For class-conditional generation the label enters as $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y)$ and is carried unchanged below.

Sampling. Generation reverses the chain from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, each step using $\boldsymbol{\epsilon}_\theta$ to remove noise. The original DDPM sampler makes this step stochastic. It needs many steps and injects fresh noise at each one, so the same $\mathbf{x}_T$ decodes to a different image on every run. This per-step noise is optional. Dropping it leaves the sample distribution intact and recasts generation as integrating the continuous-time *probability-flow ODE* [21], whose deterministic sampler was introduced by DDIM [20] and accelerated by fast solvers such as DPM-Solver [15]. No randomness then survives past $\mathbf{x}_T$, and the sampler becomes a fixed map

$$\mathbf{x}_0 = G_\theta(\mathbf{x}_T, y). \tag{4}$$

The generated image is thus a deterministic function of the initial noise alone, so $\mathbf{x}_T$ by itself determines its quality and improving a generation reduces to choosing a better $\mathbf{x}_T$.

2.2 Inference-time scaling as search over initial noise

Inference-time scaling trades extra compute at generation time for sample quality, without retraining the model. The idea is established in language modelling, where sampling many candidate answers and returning the one a verifier prefers improves accuracy predictably with the number of candidates [2, 3, 19, 13].

The same idea transfers to diffusion, but with a twist specific to it [16]. Once the solver budget is moderate the first axis is spent, and extra denoising steps leave quality essentially flat. The axis that remains open is the initial noise. By Equation (4) the image is fixed by $\mathbf{x}_T$, and the draws of $\mathcal{N}(\mathbf{0}, \mathbf{I})$ are not interchangeable. Some decode to clean, on-distribution images and others to artefacts. Inference-time scaling for diffusion is therefore a search for good initial noise,

$$\mathbf{x}_T^\star = \arg\max_{\mathbf{x}_T} \; V(G_\theta(\mathbf{x}_T, y),\, y)\,, \tag{5}$$

in which a *verifier* V scores a finished generation and the search returns the best-scoring latent. Two search strategies recur.

Random search is the analogue of best-of-N.

1. Draw n independent latents $\mathbf{x}_T^{(1)}, \ldots, \mathbf{x}_T^{(n)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. Render and score each one, $V\Big(G_\theta(\mathbf{x}_T^{(i)}, y),\, y\Big)$.
3. Keep the highest-scoring latent.

Zero-order search is a gradient-free hill-climb that refines a single latent instead of sampling globally.

1. Initialise a pivot $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. Propose N neighbours in a small ball around the pivot.
3. Render and score each neighbour, and move the pivot to the best-scoring one.
4. Repeat the previous two steps for K rounds.

The verifier bottleneck. Everything above rests on V, and in the original formulation V is always an *external* model such as CLIPScore, an aesthetic predictor, or a human-preference reward model [16]. This dependence has two drawbacks. First, a verifier must be selected or trained for the metric one cares about, which presumes that metric is known and reducible to a learnable score. Second, the search optimises the verifier's proxy rather than true sample quality, so any bias in the verifier is amplified into the generations, an effect reported as verifier hacking [16]. A score that is internal to the diffusion model, requires no preference data, and commits to no fixed notion of quality would remove both drawbacks at once. The next section presents the candidate this thesis builds on.

2.3 Generative uncertainty

The motivating intuition is that a confident generation should be robust to small perturbations of the pipeline that produced it. If a latent has settled firmly into one mode of the learned distribution, perturbing the model slightly should return the same image. If it sits on a boundary between modes, the same perturbation should send it elsewhere. The disagreement among perturbed generations is thus a usable signal of how trustworthy a sample is, and a Bayesian treatment of the network weights turns it into a per-sample score [9].

A posterior over weights. Bayesian deep learning replaces a single trained weight vector $\hat{\theta}$ with a posterior $q(\theta \mid \mathcal{D})$ that expresses which weights are plausible given the data. Computing this posterior exactly is infeasible at diffusion scale, so it is approximated post hoc by a last-layer Laplace approximation [4, 11], a Gaussian centred at $\hat{\theta}$ with covariance set by the local curvature of the loss,

$$q(\theta \mid \mathcal{D}) = \mathcal{N}\Big(\theta \mid \hat{\theta},\, \boldsymbol{\Sigma}\Big)\,, \qquad \boldsymbol{\Sigma} = \big(\nabla_\theta^2 \mathcal{L}(\theta; \mathcal{D})\big|_{\hat{\theta}}\big)^{-1}. \tag{6}$$

Restricting the approximation to the final layer keeps it tractable while still letting the weights that most directly shape the output vary.

From weight posterior to a per-sample score. The posterior turns this intuition into a measurable quantity. Fix an initial noise $\mathbf{x}_T$ and ask whether plausible models agree on what it should become. Draw M weight samples $\theta_m \sim q(\theta \mid \mathcal{D})$ and decode the *same* latent under each, $\hat{\mathbf{x}}^{(m)} = G_{\theta_m}(\mathbf{x}_T, y)$. The completions must be compared by content rather than by pixels, since two images of the same object can differ in every pixel, so each is mapped to a semantic vector by a frozen CLIP image encoder c_φ [17],

$$\mathbf{e}_m = c_\varphi\Big(\hat{\mathbf{x}}^{(m)}\Big) \in \mathbb{R}^D, \qquad m = 1, \ldots, M. \tag{7}$$

Their spread in this feature space is summarised by moment-matching the M embeddings to a single diagonal Gaussian, the semantic posterior predictive,

$$p(\mathbf{x} \mid \mathbf{x}_T, \mathcal{D}) \approx \mathcal{N}\Big(\mathbf{e}(\mathbf{x}) \,\Big|\, \bar{\mathbf{e}},\, \mathrm{Diag}\Big(\tfrac{1}{M}\textstyle\sum_m \mathbf{e}_m^2 - \bar{\mathbf{e}}^2\Big) + \sigma^2\mathbf{I}\Big), \qquad \bar{\mathbf{e}} = \tfrac{1}{M}\displaystyle\sum_{m=1}^{M} \mathbf{e}_m, \tag{8}$$

where σ^2 is the observation noise of the semantic likelihood. *Generative uncertainty* is the entropy of this Gaussian,

$$u(\mathbf{x}_T) = H\big(p(\mathbf{x} \mid \mathbf{x}_T, \mathcal{D})\big). \tag{9}$$

Entropy grows with the embedding variance, so a low $u(\mathbf{x}_T)$ marks a stable generation on which the perturbed models concur, and a high $u(\mathbf{x}_T)$ a fragile one that small perturbations push to semantically different images. The score ranks the sample from the unperturbed run, with the M perturbed completions used only to estimate its spread.

3 Method

With the insights of Section 2 in place, we now present our framework for verifier-free inference-time scaling. Two questions must be answered: how to score an initial noise using only the diffusion model itself (Section 3.1), and how to search for better noise under that score (Section 3.2). Section 3.3 instantiates the external CLIP verifier against which the framework is judged.

3.1 Generative uncertainty from input perturbations

Both estimators in this section compute the same score (Equation (9)): for a fixed initial noise $\mathbf{x}_T$, produce M perturbed completions, embed them with the CLIP encoder c_φ, and take the entropy over those embeddings. In both cases the kept image is the *unperturbed* completion $G_{\theta_{\text{MAP}}}(\mathbf{x}_T, y)$, and the M perturbed completions serve only to score it. The estimators differ in one place only, namely where the perturbation enters.

The Laplace baseline. The baseline is the original estimator of Section 2.3 [9]: M weight vectors $\theta_m \sim \mathcal{N}(\theta_{\text{MAP}}, \boldsymbol{\Sigma})$ drawn from the diagonal last-layer Laplace posterior (Equation (6)), each completing the same latent,

$$\hat{\mathbf{x}}^{(m)} = G_{\theta_m}(\mathbf{x}_T, y), \qquad m = 1, \ldots, M. \tag{10}$$

Input perturbations instead of Laplace. We keep the weights fixed at θ_{MAP} and perturb the initial noise instead. Each scoring completion starts from a perturbed latent

$$\mathbf{x}_T^{(m)} = \frac{\mathbf{x}_T + \sigma_{\text{p}}\,\boldsymbol{\epsilon}^{(m)}}{\sqrt{1+\sigma_{\text{p}}^2}}, \qquad \boldsymbol{\epsilon}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad m = 1, \ldots, M, \tag{11}$$

and is completed under the shared weights, $\hat{\mathbf{x}}^{(m)} = G_{\theta_{\text{MAP}}}(\mathbf{x}_T^{(m)}, y)$. The normalisation by $\sqrt{1+\sigma_{\text{p}}^2}$ is variance-preserving. Since $\mathbf{x}_T$ and $\boldsymbol{\epsilon}^{(m)}$ are independent draws of $\mathcal{N}(\mathbf{0}, \mathbf{I})$, each $\mathbf{x}_T^{(m)}$ again has the $\mathcal{N}(\mathbf{0}, \mathbf{I})$ marginal that the sampler was trained to expect, so the perturbed completions stay on the same prior as the base sample and σ_{p} controls only how far they are nudged.

The change is small in the estimator but decisive in cost. Because all M completions in Equation (11) share the weights θ_{MAP}, they form a single batch of M latents and are produced in *one* batched forward pass rather than M sequential ones. The perturbed completions ($m = 1, \ldots, M$) may additionally be rendered on a shorter denoising schedule than the kept image ($m = 0$), since they only contribute to the score (Equation (9)) of the kept image. Algorithm 1 summarises the estimator.

Algorithm 1: Generative-uncertainty score via input perturbation

Input : noise $\mathbf{x}_T$, class y, fixed model G_θ, samples M, perturbation scale σ_{p}, encoder c_φ, observation noise σ^2
Output : kept sample $\hat{\mathbf{x}}_0$, uncertainty $u(\mathbf{x}_T)$

$\hat{\mathbf{x}}_0 \leftarrow G_{\theta_{\mathrm{MAP}}}(\mathbf{x}_T, y)$ `// unperturbed sample to keep`
for $m = 1$ **to** M **do**
  $\boldsymbol{\epsilon}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\mathbf{x}_T^{(m)} \leftarrow (\mathbf{x}_T + \sigma_{\mathrm{p}}\, \boldsymbol{\epsilon}^{(m)}) / \sqrt{1 + \sigma_{\mathrm{p}}^2}$ `// Equation (11)`
  $\mathbf{e}_m \leftarrow c_\varphi\big(G_{\theta_{\mathrm{MAP}}}(\mathbf{x}_T^{(m)}, y)\big)$ `// batched over` m
$\bar{\mathbf{e}} \leftarrow \frac{1}{M} \sum_m \mathbf{e}_m$
$u(\mathbf{x}_T) \leftarrow H\big(\mathcal{N}(\bar{\mathbf{e}},\, \mathrm{Diag}(\frac{1}{M} \sum_m \mathbf{e}_m^2 - \bar{\mathbf{e}}^2) + \sigma^2 \mathbf{I})\big)$ `// Equations (8) and (9)`
return $\hat{\mathbf{x}}_0,\ u(\mathbf{x}_T)$

3.2 Zero-order search with generative uncertainty as verifier

We now use generative uncertainty as a search *objective* rather than a filter, and actively move the initial noise toward latents of low generative uncertainty through the zero-order search introduced in Section 2.2 [16]. The search maximises a generic score $s(\mathbf{x}_T, y)$, the role the verifier V plays in Equation (5).

Starting from a random pivot, each of K iterations proposes N neighbours through the variance-preserving step

$$\mathbf{x}'_T = \sqrt{1 - \lambda^2}\, \mathbf{x}_T + \lambda\, \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{12}$$

which keeps the $\mathcal{N}(\mathbf{0}, \mathbf{I})$ marginal while λ sets the neighbourhood radius. Each neighbour is scored with the input-perturbation estimator of Section 3.1, and the pivot moves to the best-scoring candidate. The current pivot is included among the candidates, so the pivot score cannot worsen across iterations. After K iterations the final pivot is rendered once at the full denoising schedule to produce the output image.

Algorithm 2: Zero-order search over initial noise

Input : class y, score $s(\cdot, y)$, iterations K, neighbours N, radius λ, generator G_θ
Output : generated image $\mathbf{x}_0$
$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ `// random pivot`
for $k = 1$ **to** K **do**
  for $n = 1$ **to** N **do**
    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
    $\mathbf{x}_T^{(n)} \leftarrow \sqrt{1 - \lambda^2}\, \mathbf{x}_T + \lambda\, \boldsymbol{\epsilon}$ `// Equation (12)`
  $\mathbf{x}_T \leftarrow \arg\max_{\mathbf{c} \in \{\mathbf{x}_T, \mathbf{x}_T^{(1)}, \ldots, \mathbf{x}_T^{(N)}\}} s(\mathbf{c}, y)$ `// pivot kept among candidates`
return $G_\theta(\mathbf{x}_T, y)$ `// render at full schedule`

3.3 Zero-order search with an external CLIP verifier

To judge generative uncertainty we compare it against the external-verifier search it is meant to replace. We hold the search procedure of Section 3.2 fixed, the same zero-order hill-climb over $\mathbf{x}_T$, and the same K and N, and change only the score $s(\mathbf{x}_T, y)$ that ranks candidates. Where generative uncertainty scores a candidate by the *internal* disagreement of M perturbed completions, an external verifier scores the *single* completion $G_\theta(\mathbf{x}_T, y)$ with a separate frozen model. We use the CLIP zero-shot classifier [17] as the verifier, in two variants that differ only in whether the score is allowed to read the target label y.[1]

The shared zero-shot classifier. Both variants build on the CLIP zero-shot classifier, and we follow the prompt engineering of Radford et al. [17]: each ImageNet class c is turned into a text embedding $\mathbf{w}_c \in \mathbb{R}^D$ by averaging the CLIP text features over 80 prompt templates (e.g. "a photo of a {class}.") and renormalising. $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_{1000}] \in \mathbb{R}^{D \times 1000}$. A candidate's rendered image is embedded with the *same* frozen CLIP image encoder c_φ used by the

[1] We use only the CLIP classifier so that the verifier and the generative-uncertainty score share one image encoder, isolating the choice of score from the choice of feature extractor.

generative-uncertainty score (Section 3.1),

$$\hat{\mathbf{e}}(\mathbf{x}_T) = \frac{c_\varphi(G_\theta(\mathbf{x}_T, y))}{\left\| c_\varphi(G_\theta(\mathbf{x}_T, y)) \right\|}, \tag{13}$$

so that $\langle \hat{\mathbf{e}}(\mathbf{x}_T), \mathbf{w}_c \rangle$ is the cosine similarity between the generation and class c.

Labelled verifier (CLIP zero-shot classifier). The first variant is the external classifier verifier used for class-conditional ImageNet [16]. We build the CLIP zero-shot classifier once, so every class has its text vector $\mathbf{w}_c$ as a column of $\mathbf{W}$. To score a candidate we render its image, embed it with CLIP into $\hat{\mathbf{e}}(\mathbf{x}_T)$, and take the cosine similarity to the vector of the *target* class y,

$$s_{\text{labelled}}(\mathbf{x}_T, y) = \big\langle \hat{\mathbf{e}}(\mathbf{x}_T),\, \mathbf{w}_y \big\rangle. \tag{14}$$

A high s_{labelled} means CLIP reads the generation as clearly class y. The search compares this score across the candidates and moves the pivot to the highest. This is a strong but deliberately privileged baseline. It reads the same label y the diffusion model is conditioned on, and rewards exactly the property the downstream metric cares about, that a sample generated for class y is recognisable as class y. It marks the ceiling for what external-verifier search can buy in this setting.

Label-free verifier (CLIP confidence). The labelled verifier has an advantage generative uncertainty lacks. It is handed the target class y, while generative uncertainty never sees y. A gap between them would then be ambiguous, reflecting either a better signal or simply more information. The second variant removes this asymmetry. Instead of comparing the image to one known class, it runs the candidate through the whole classifier to get CLIP's probability over all 1000 classes,

$$\mathbf{p}(\mathbf{x}_T) = \operatorname{softmax}\big(\tau\, \mathbf{W}^\top \hat{\mathbf{e}}(\mathbf{x}_T)\big), \tag{15}$$

where τ is CLIP's logit scale. The candidate is then scored by the *negative entropy* of this posterior,

$$s_{\text{labelfree}}(\mathbf{x}_T) = -H\big(\mathbf{p}(\mathbf{x}_T)\big) = \sum_{c=1}^{1000} p_c(\mathbf{x}_T)\, \log p_c(\mathbf{x}_T). \tag{16}$$

This is largest when the posterior is peaked on a single class and lowest if uniform. A high $s_{\text{labelfree}}$ thus marks a generation CLIP confidently assigns to one class, whichever it is. The score never uses y, and the search moves the pivot to the highest-scoring candidate.

4 Results

We report results for the two research questions of Section 1. Section 4.1 (RQ1) tests whether input-noise perturbation can replace the last-layer Laplace posterior as the source of perturbations in generative uncertainty, used as a post-hoc filter. Section 4.2 (RQ2) tests whether the input perturbation version of generative uncertainty, used as the objective of a zero-order search over initial noise, is competitive with an external CLIP verifier at matched compute.

Dataset. We study class-conditional generation on ImageNet-1k at 256×256 [5], with class labels drawn uniformly over the 1000 classes. FID and Precision/Recall are computed against a reference set of $50{,}000$ real ImageNet images. The last-layer Laplace posterior of RQ1 is fit on a $1/50$ subsample of the training split ($25{,}623$ images).

Models. The generator is a pretrained U-ViT-H/2 backbone [1], a 28-layer transformer operating in the latent space of a Stable Diffusion KL autoencoder [18], with weights frozen throughout. We sample with the second-order DPM-Solver [15] on a time-uniform schedule.

Verifiers. RQ1 compares two generative-uncertainty estimators as filters, the last-layer Laplace score [9] and our input-perturbation score, both with $M{=}5$ perturbed members and input perturbation at $\sigma_{\text{p}}{=}0.125$. RQ2 uses the input-perturbation score as a search objective and compares it against two external verifiers built on a frozen CLIP ViT-B/32 [17]: the labelled zero-shot classifier, which reads the target class, and the label-free CLIP-confidence score, which does not.

Metrics. Quality is measured by FID [7] and Improved Precision and Recall [12], both in Inception-v3 feature space [22]. FID is the distance between the generated and real feature distributions, lower being closer, but it conflates fidelity and coverage. Precision, the fraction of generated samples that fall in the region covered by the real data, measures fidelity. Recall, the fraction of real samples that fall in the region covered by the generated data, measures coverage.

4.1 Generative uncertainty: input perturbation versus last-layer Laplace

RQ1 asks whether input-noise perturbation can replace the Laplace weight-space posterior as the filtering signal. We score a fixed pool of $12{,}032$ samples by generative uncertainty and retain the n of lowest uncertainty, sweeping n from

6K to the full pool. Both estimators score the *same* pool, identical initial noises and labels, so the comparison is paired. Because FID improves with sample count, each filter is judged against a random baseline retaining the same n. We test the Laplace score, the input-perturbation score at the full schedule (T=50), and the input-perturbation score at cheaper screening schedules ($T \in \{25, 10, 5\}$).

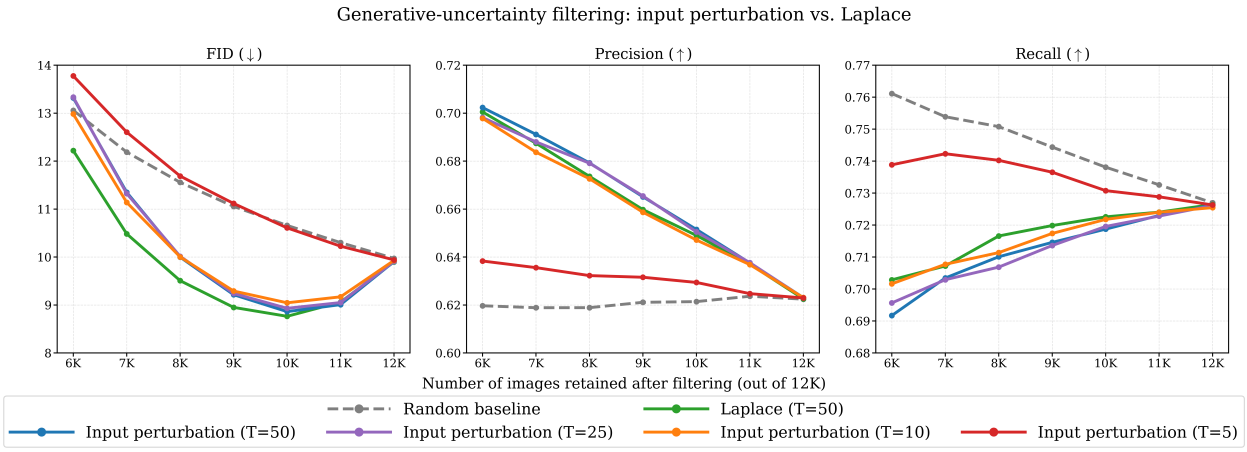


Figure 1: **RQ1: generative-uncertainty filtering, input perturbation versus last-layer Laplace.** Each method scores a fixed pool of 12,032 ImageNet 256 × 256 samples and retains the n images of lowest uncertainty. The random baseline retains an unfiltered subset of the same size. Panels: FID (↓), Precision (↑), Recall (↑) against retained count n. Curves: random baseline, the Laplace filter at the base schedule (T=50), and the input-perturbation filter at the full base schedule (T=50) and three cheaper screening schedules ($T \in \{25, 10, 5\}$ denoising steps). All filters use M=5 perturbed members and σ_{p}=0.125. Because FID decreases with sample count, the fair comparison at each n is a filter against the random baseline at the same n.

As Figure 1 shows, the input-perturbation filter at the full schedule beats random at every retention level except the strictest, and reaches its lowest FID at N=10K: discarding the most uncertain ≈17% of the pool gives a FID of 8.86 against 10.66 for random at the same count (N=10K). There it nearly matches the Laplace reference (8.86 against 8.76). Laplace and input perturbation separate only in the most-confident tail, where Laplace ranks more sharply (12.22 against 13.32 at N=6K), and the gap closes as the filter relaxes, nearly vanishing by N=10K. Pushing either filter into that tail is counter-productive regardless: at N=6K the sample-count penalty on FID erases the gain.

Precision and Recall show where the FID gain comes from. Filtering trades coverage for fidelity: at N=10K the input-perturbation filter raises Precision from 0.621 (random) to 0.652 while lowering Recall from 0.738 to 0.719. Retaining only confident samples removes broken generations, but the score cannot distinguish rare-and-broken from rare-and-valid. So unusual yet realistic images are discarded with them, therefore lowering the recall. On both metrics the Laplace and input-perturbation filters trace nearly coincident curves across the sweep, which supports the substitution beyond the FID panel.

Because the perturbed members only rank the kept image, they can use a shorter schedule. Both T=25 and T=10 track the full schedule closely across the sweep, so the score survives down to a fifth of the base schedule. Only T=5 collapses to the random baseline.

In sum, input-noise perturbation reproduces the Laplace filter at the FID optimum and stays robust down to a fifth of the denoising schedule, turning a score of M sequential trajectories into one batched, short-schedule pass. This answers RQ1 affirmatively, with one caveat, the strict-filter tail, where Laplace still ranks more sharply.

4.2 Zero-order search: generative uncertainty versus external verifiers

RQ2 asks whether the generative-uncertainty score, used as the objective of a zero-order search over initial noise (Section 3.2), is competitive with an external CLIP verifier (Section 3.3). We compare three searches that share the same zero-order procedure (K=3 iterations, N=2 neighbours, λ=0.2) and differ only in the score: generative uncertainty, the labelled CLIP verifier s_{labelled} which reads the target class, and the label-free CLIP-confidence verifier $s_{\text{labelfree}}$ which does not.

All three searches improve sharply over unguided sampling. The unguided reference is the random baseline of Figure 1 at the same sample count (n=10K) and scores FID 10.66. Every search cuts this by roughly a quarter or more: the labelled CLIP verifier reaches 6.95, the label-free verifier 7.26, and generative uncertainty 7.99 (Figure 2).

The searches show the same trade as filtering: all three raise Precision and lower Recall relative to unguided sampling. Generative uncertainty keeps the most diversity (Recall 0.713) but gains the least fidelity (Precision 0.674). The label-free verifier sits at the opposite end and the labelled verifier in between. Generative uncertainty trails both verifiers in Precision and in FID. The two CLIP verifiers are nearly tied on both metrics. The overall pattern matches what each score rewards. CLIP confidence is highest for textbook examples of a class, so its search steers candidates toward such images. These images are realistic but similar to one another, which raises Precision and costs Recall. The uncertainty score only asks whether an image survives small nudges, not what it shows. An unusual but stable image scores as well as a textbook one, so variety survives and Precision gains less.

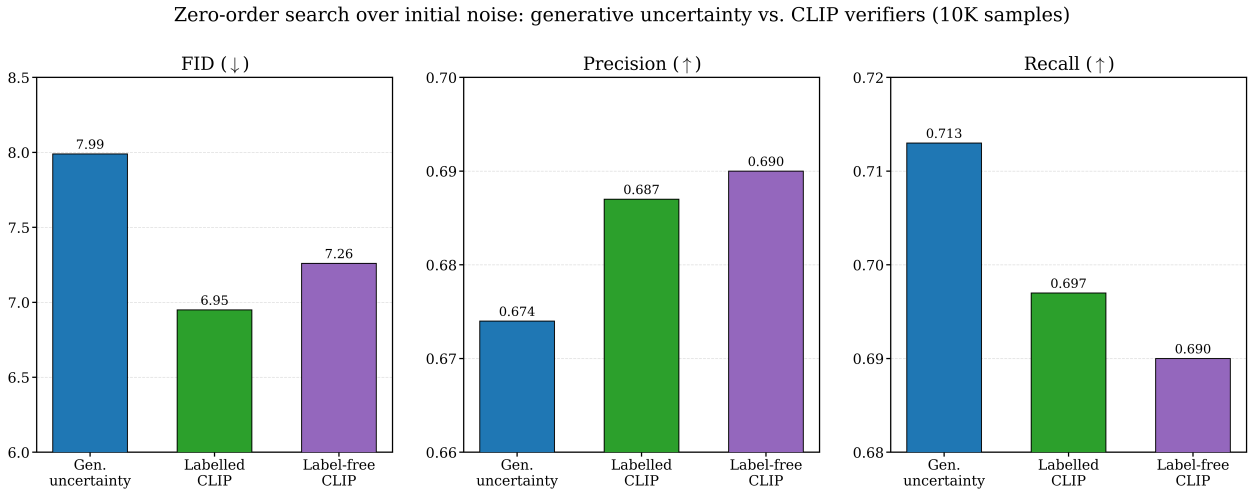


Figure 2: **RQ2: zero-order search over initial noise.** FID (↓), Precision (↑) and Recall (↑) over 10,000 generated images for three search scores at equal search depth (K=3, N=2, λ=0.2) and matched compute.

As a search objective generative uncertainty therefore works in absolute terms, improving substantially over unguided sampling, but it does not rival an external CLIP verifier. At equal search depth and compute it is outperformed by both the labelled and the label-free verifier, so on this benchmark the model's internal self-consistency is a weaker search signal than CLIP's class confidence.

4.3 Qualitative samples

Figure 3 shows the two ends of the ranking behind these numbers: the lowest- and highest-uncertainty generations of the RQ1 pool. The low-uncertainty end is dominated by single, centred subjects in canonical poses against uncluttered backgrounds. The high-uncertainty end collects the generator's failure modes, distorted faces and hands, garbled text, and crowded multi-object scenes. The score evidently flags broken generations, but the same preference penalises complex compositions in general; this is the qualitative face of the Recall cost measured in Figures 1 and 2.



Figure 3: **What the score selects.** The 25 lowest-uncertainty (left) and 25 highest-uncertainty (right) generations in the RQ1 pool of 12,032 ImageNet 256×256 samples, ranked by the input-perturbation score (M=5, σ_{p}=0.125, T=50). Low uncertainty selects single, centred subjects in canonical poses; high uncertainty concentrates on distorted faces, garbled text, and cluttered scenes.

Figure 4 makes the mechanism visible. For three classes we take the most and the least certain latent and show the kept base image next to its M=5 perturbed completions. Where uncertainty is low, the completions are semantically indistinguishable from the base: the same retriever, the same mountain, the same burger, differing only in texture. Where uncertainty is high, the identical perturbation scale scatters the completions: the retriever changes pose, scene and framing, the alp dissolves into incoherent shapes, the cheeseburger drifts toward a different dish entirely. The entropy of Equation (9) is a measure of exactly this divergence.

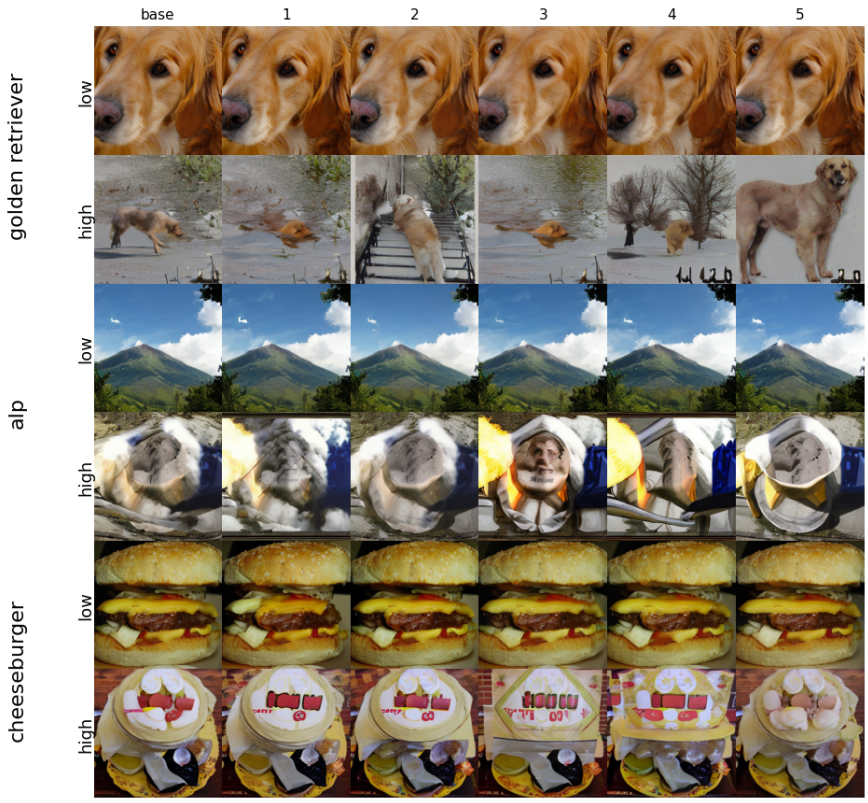


Figure 4: **Perturbation ensembles behind the score.** For three ImageNet classes (golden retriever, alp, cheeseburger), the most certain (*low*) and least certain (*high*) latent among the generations of that class. Each row shows the kept base image (m=0, schedule T=50) followed by its M=5 perturbed completions (Equation (11), σ_{p}=0.125). Low-uncertainty latents yield completions that agree semantically with the base; high-uncertainty latents scatter under the same perturbation scale. This disagreement is what the score $u(\mathbf{x}_T)$ of Equation (9) measures.

5 Discussion & Limitations

Why the substitution works. RQ1 shows that perturbing the input reproduces the behaviour of perturbing the weights. This is less surprising than it first appears. Both estimators test how stable the map $G_\theta(\mathbf{x}_T, y)$ is around the same point. One nudges the weights θ and the other nudges the noise $\mathbf{x}_T$. A latent in a stable region of the map tolerates either nudge. A latent near a boundary tolerates neither. The estimators separate only in the most-confident tail (n=6K in Figure 1), where Laplace ranks more sharply. We hypothesise that the weight posterior helps there because its noise is shaped by the training data. By Equation (6) the Laplace covariance is the inverse curvature of the loss, so it is larger in directions the data leaves uncertain, while input noise is equally strong in every direction. Testing this would need structured input noise, and we leave it open.

What the search optimises, and why CLIP wins. RQ2 has a two-sided outcome. Generative uncertainty works as a search objective, yet it is the weaker one at matched compute. The Precision and Recall panels say why. Self-consistency rewards images the model renders stably. Stability correlates with fidelity but says nothing about the class, so the search keeps diversity (highest Recall, 0.713) while gaining the least Precision (0.674). CLIP rewards close to what the metrics measure. FID, Precision and Recall are all computed in the feature space of an ImageNet classifier [22], and CLIP confidence is itself a classifier-style score, so part of its advantage may be agreement with the measuring stick rather than truly better images. Our results cannot rule this out. Settling it would need metrics that do not rest on a classifier, or human judgement.

Verifier hacking, traded rather than removed. Removing the external verifier removes the risk of optimising another model's biases [16], but it does not remove bias. The uncertainty score prefers single, centred subjects and penalises complex scenes wholesale (Figure 3), so searching against it drifts toward simple imagery. This is a self-imposed form of verifier hacking, and it costs diversity: Recall falls from 0.738 (unguided) to 0.713 under the search. The CLIP searches lose even more Recall, since CLIP penalises complex scenes and unusual views alike. The uncertainty search

therefore pays the smallest diversity cost of the three, but the cost is not zero. Verifier-free does not mean bias-free. Where coverage of the data distribution matters, the smallest diversity loss is still an argument for the uncertainty objective despite its worse FID.

Limitations. All results come from single runs at one fixed seed, and no confidence intervals are reported. FID differences as small as the one between input perturbation and Laplace (8.86 vs. 8.76 at n=10K) should be read as no evidence of a difference rather than as equality. The NFE accounting excludes autoencoder decoding and CLIP encoding, and these excluded costs are not symmetric. The uncertainty score decodes and encodes M=5 images per candidate where a verifier handles one. The hyperparameters (σ_{p}=0.125, M=5, K=3, N=2, λ=0.2) were fixed for all experiments without real proper tuning. The search was run at a single shallow depth, so how either objective scales with larger budgets is unmeasured.

6 Conclusion

This thesis asked whether a diffusion model's own generative uncertainty can replace an external verifier in inference-time scaling, both as a score for selecting samples and as an objective for searching over initial noise.

RQ1 asked whether input-noise perturbations can replace the Laplace posterior as the source of perturbations in generative uncertainty. Yes: filtering a fixed pool by the input-perturbation score retains samples with FID matching the Laplace filter at the operating optimum (8.86 vs. 8.76, against 10.66 unfiltered) and nearly coincident Precision and Recall across the sweep, with Laplace retaining an edge only under the strictest filtering. The score, moreover, survives screening at a fifth of the denoising schedule. The substitution turns M sequential trajectories into one batched, short-schedule pass, making the score cheap enough to search with.

RQ2 asked whether this score can serve as the verifier inside a zero-order search over initial noise, and how it compares to an external CLIP verifier at matched compute. As a search objective it works: at 400 NFE per image it improves FID from 10.66 (unguided) to 7.99 while preserving more sample diversity than any external verifier tested. But it does not close the gap to CLIP: the labelled zero-shot classifier reaches 6.95 and the label-free confidence score 7.26 at the same budget, so the deficit is not explained by access to the target class.

The answer to the central question is therefore split. As a selection signal, generative uncertainty is a drop-in, verifier-free replacement. As a search signal, it works but does not win. It clearly improves the FID over unguided sampling, but both CLIP verifiers improve the FID considerably more at the same cost, but lose the most diversity.

7 Future work

Other verifiers. The external side of the comparison is a single family, CLIP at one model size. Classifier verifiers built on DINO features, aesthetic predictors, and reward models trained on human preferences are all established in this setting [16]. Repeating the matched-compute search with each would show whether the gap to generative uncertainty is specific to CLIP or generic to external verifiers. It would also show whether the diversity-preserving behaviour of the uncertainty objective persists against verifiers with very different biases.

Scaling the search. The search was run at one shallow depth (K=3, N=2, 400 NFE per image). This is a single point on a curve, and inference-time scaling is about the curve. Both objectives should therefore be run at more iterations and more neighbours, and against random search at the same budgets [2]. Deeper search is not a cost problem in this setting, since spending more compute per image is the stated goal. The interesting question is how the gap behaves. It may stay, widen, or close at larger depth, and single-budget results cannot tell.

Beyond class labels. In text-to-image generation there is no finite label set. Labelled verifiers do not transfer to that setting, while label-free and internal signals do. The uncertainty objective would need no change at all, since it never reads the conditioning. Whether its diversity-preserving behaviour holds there is the natural next test, and it is the setting where a verifier-free objective would matter most.

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